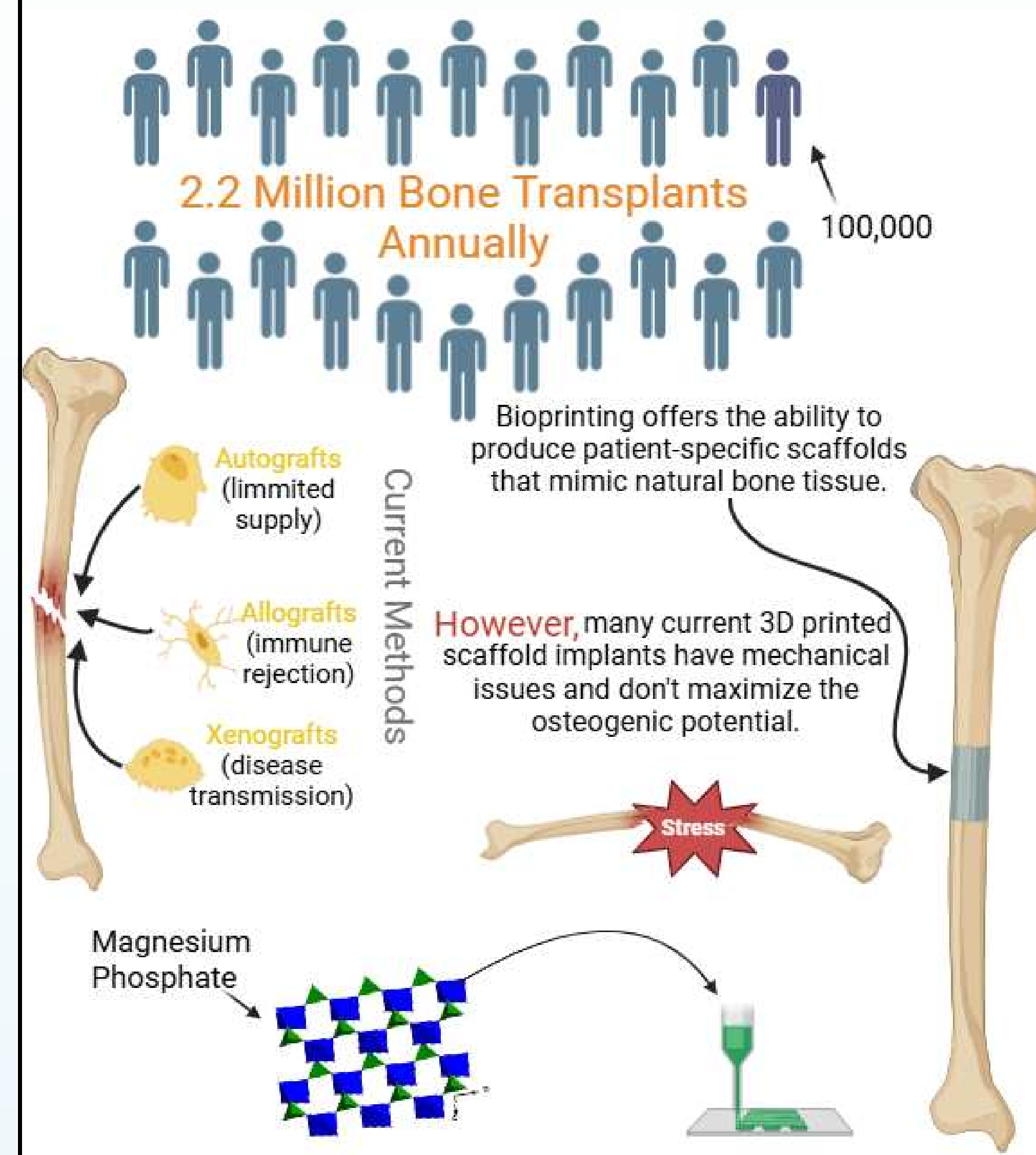


Evaluating the Effect of Differing Magnesium Phosphate Concentrations on Cellular and Mechanical Performance in MC3T3-E1 Biogel Scaffolds

Infographics Created In BioRender by Researcher • All Graphs and Data Visualization Created In Excel & Data Classroom

Background



Magnesium Phosphate and its ability to **restore the oxidative stress balance, increase expression of osteogenic proteins, and provide abundant active sites** to effectively adsorb protein molecules can potentially solve the mechanical and osteogenic issues in current scaffolds.

Purpose

To develop stronger, osteoinductive bone scaffolds capable of withstanding physiological loads for improved patient outcomes in bone tissue engineering.

Hypothesis

Increasing the concentration of magnesium phosphate ($\text{Mg}_3(\text{PO}_4)_2$) within MC3T3-E1 bioink scaffolds will **enhance their compressive stress limit while preserving cell viability**.

Variables

IV: $\text{Mg}_3(\text{PO}_4)_2$ concentration (0%, 1%, 2%)

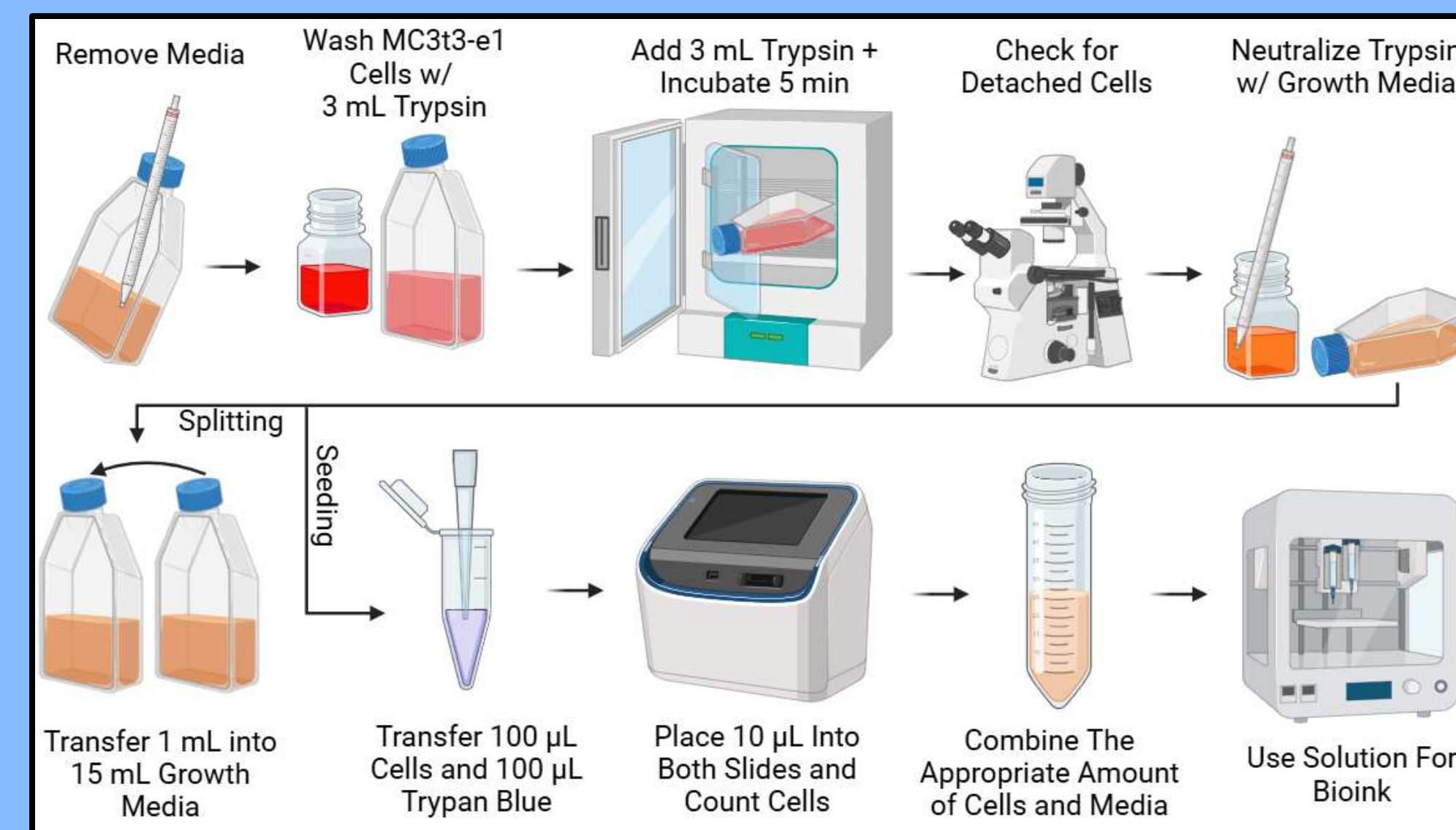
DV: 1. Cell viability

2. Elastic Limit Compressive Stress

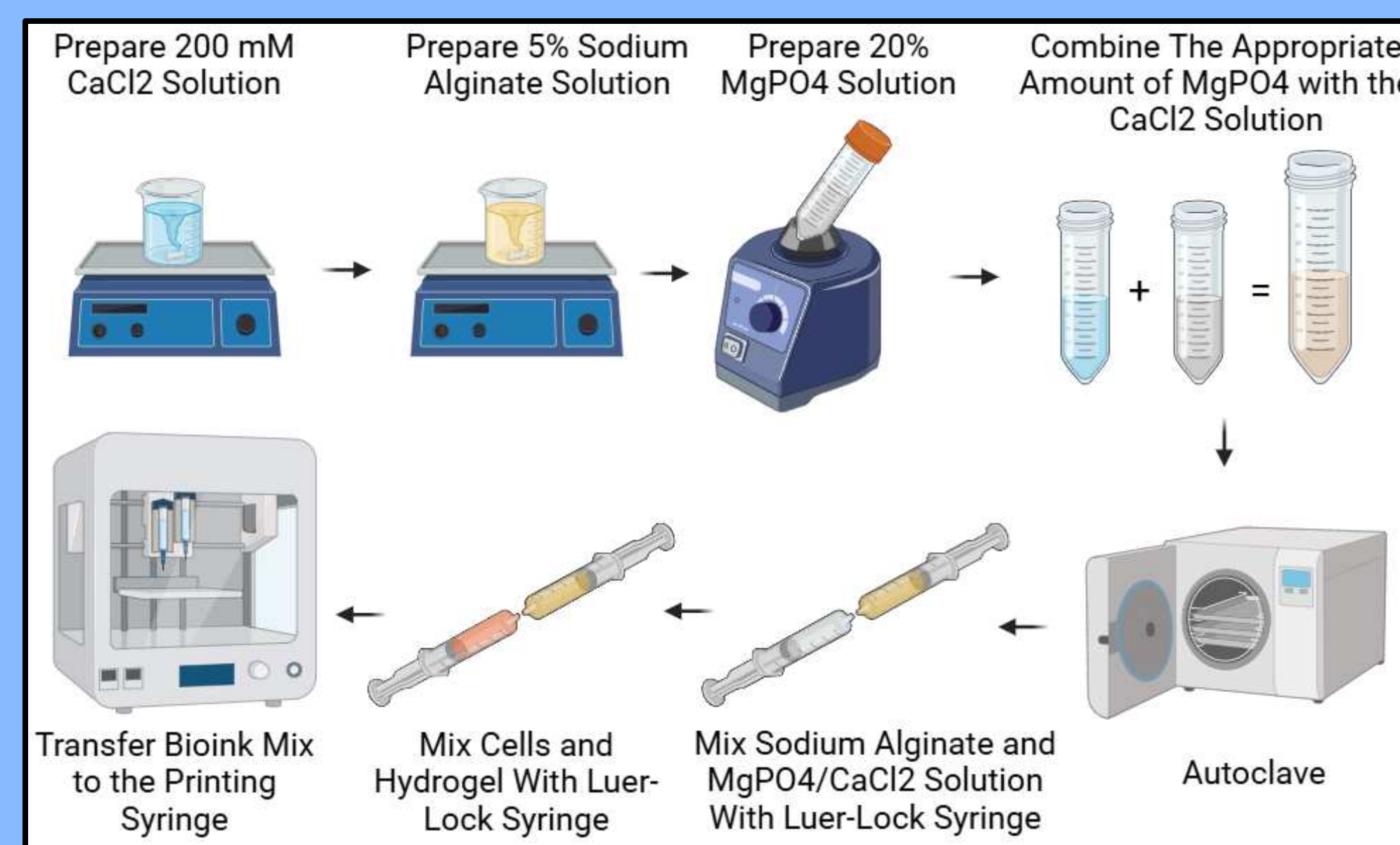
Control: Sodium-Alginate/ CaCl_2 bioink

Methodology

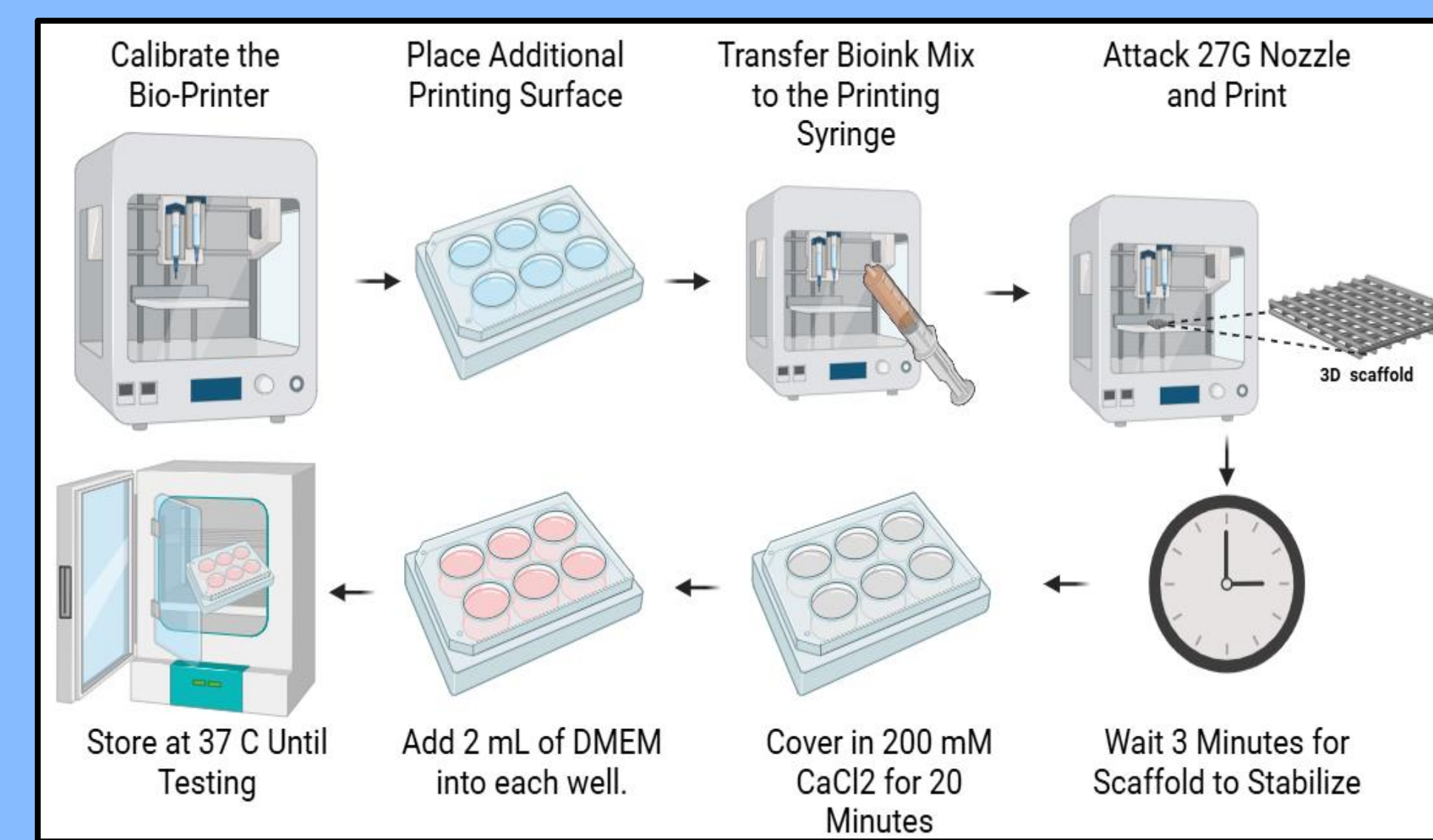
1. Split, and Seed Cells



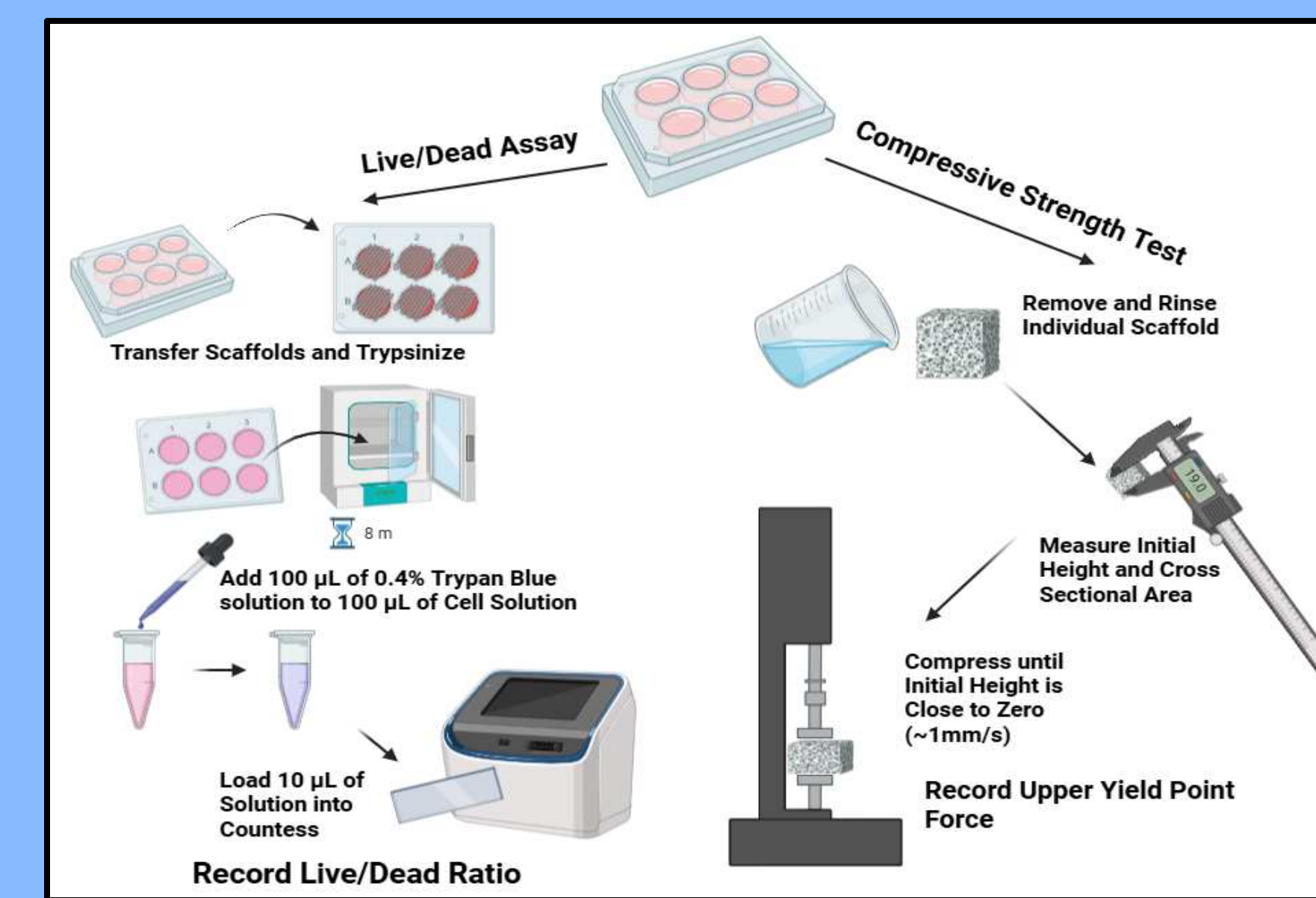
2. Make the Bioink



3. Bioprint Alginate/ $\text{CaCl}_2/\text{Mg}_3(\text{PO}_4)_2$ Scaffolds

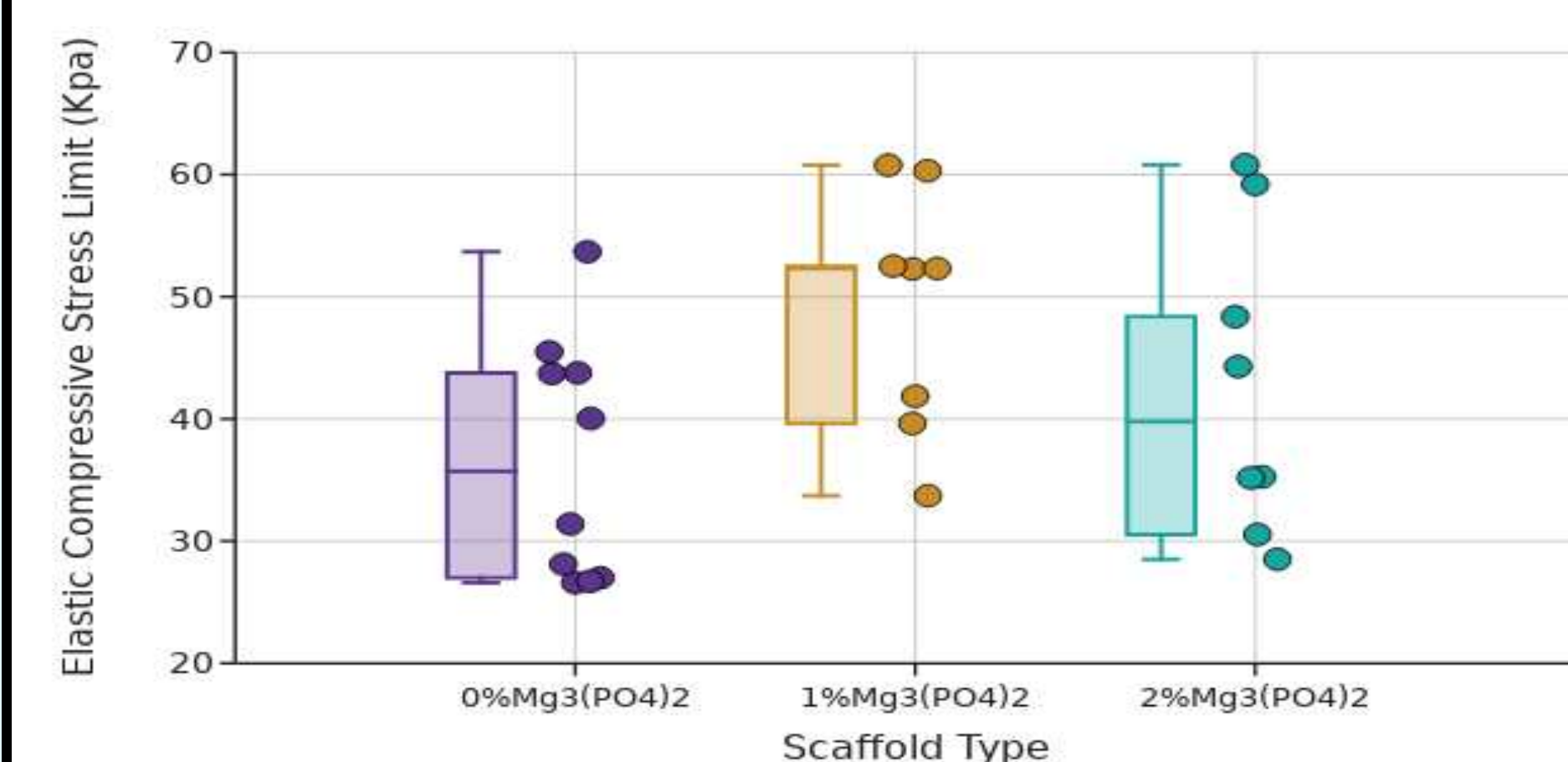


4. Cellular and Mechanical Performance



Results

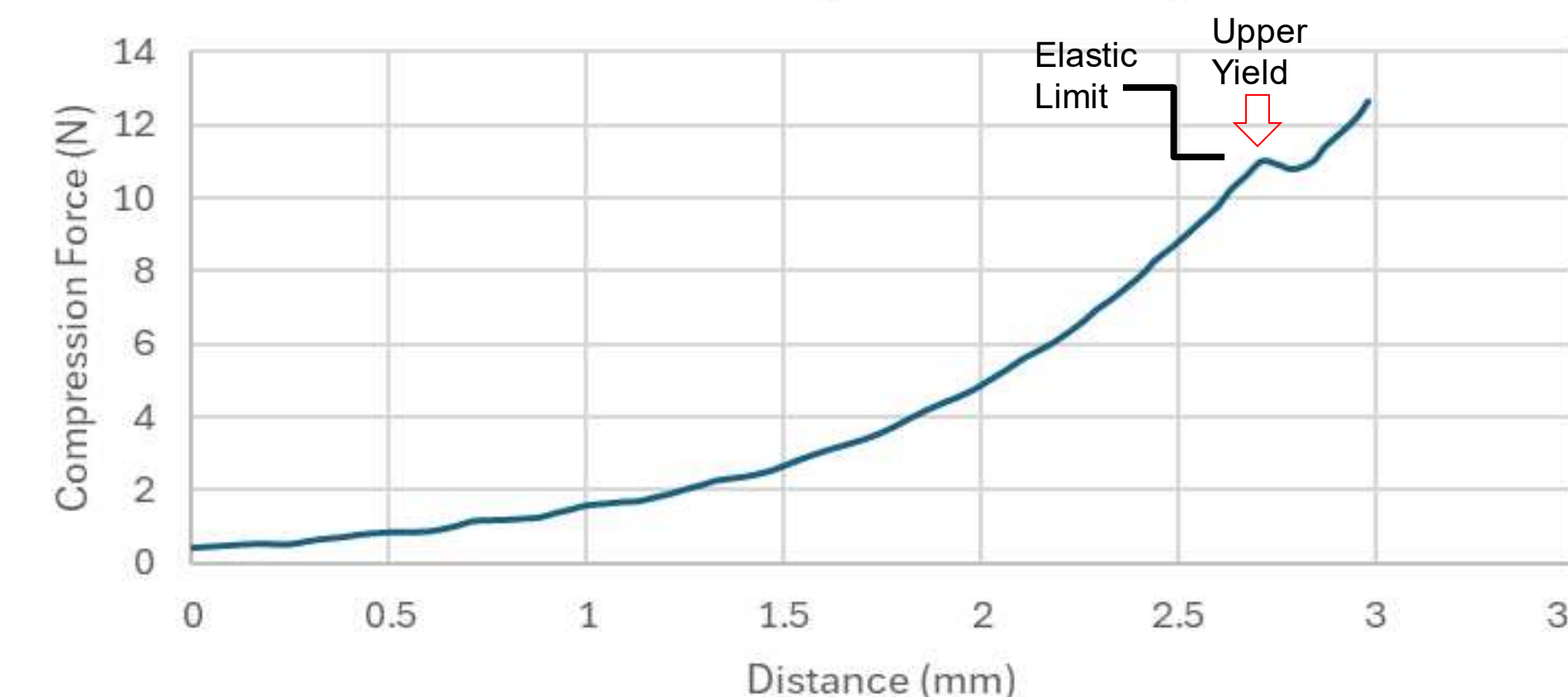
Effect of Magnesium Phosphate Concentration on Elastic Compressive Limit



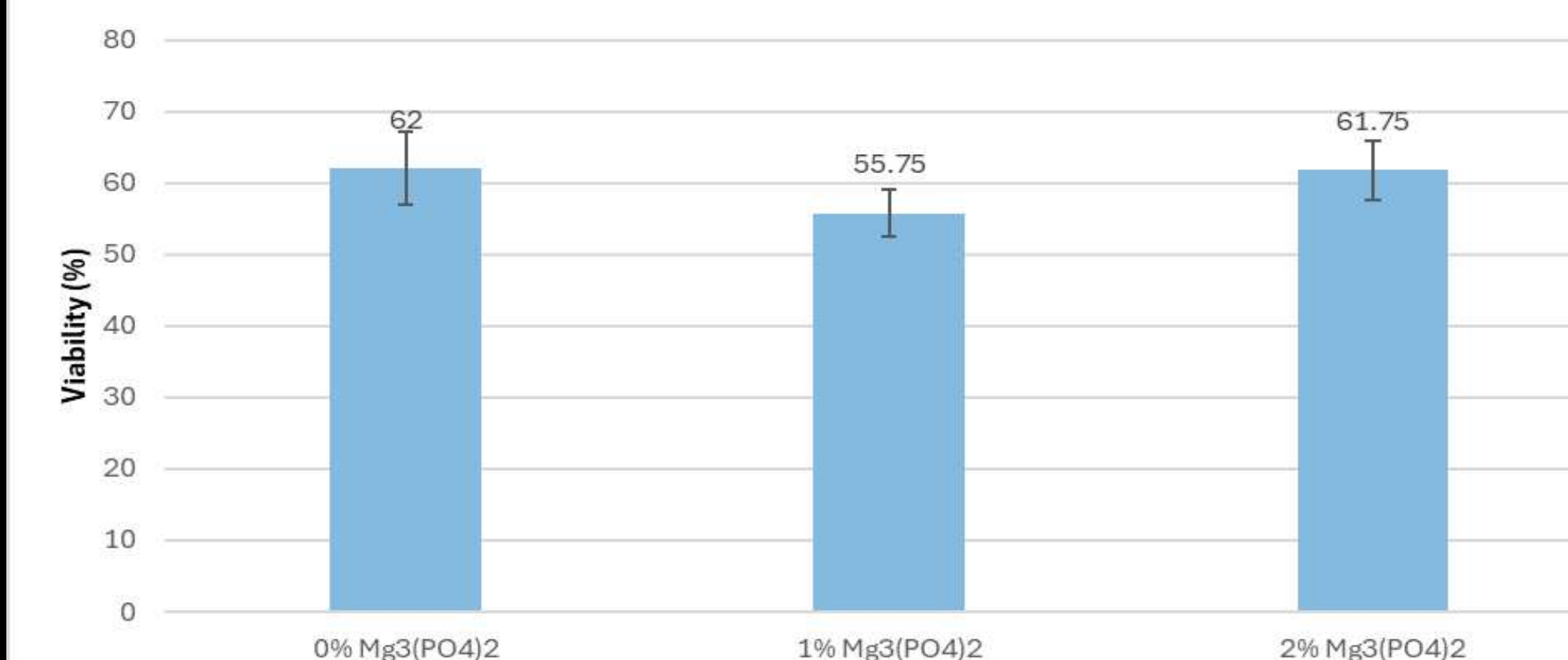
Mann-Whitney U Test Results for Pairwise Comparisons of Elastic Limit Compressive Stress Across Scaffold Types

Scaffold Pair Comparison	P-value	Significance (p < 0.05)
0% & 1% $\text{Mg}_3(\text{PO}_4)_2$	0.0455	yes
0% & 2% $\text{Mg}_3(\text{PO}_4)_2$	0.1971	no
1% & 2% $\text{Mg}_3(\text{PO}_4)_2$	0.3173	no

Example Elastic Limit Compressive Stress Test (0% Trial 1)



Viability of Sodium Alginate, Calcium Chloride, and Magnesium Phosphate Scaffolds



Discussion

Analysis

Data Supported Hypothesis –

- Significant (20.5 %) increase in the Mechanical Performance between 0% and 1% $\text{Mg}_3(\text{PO}_4)_2$.
- There was an insignificant increase between 0% and 2%, suggesting that the optimal concentration is somewhere around 1%.
- Negligible difference in Cellular Performance between all groups.

Conclusion

The addition of $\text{Mg}_3(\text{PO}_4)_2$ (~1%) to MC3T3-E1 bioink scaffolds enhances their compressive stress limit while maintaining cell viability.

Future Outlook

To further investigate the potential of $\text{Mg}_3(\text{PO}_4)_2$ enhanced scaffolds, future studies should evaluate their osteogenic and biochemical effects in more depth.

- Alizarin Red S Staining:** To assess Magnesium Phosphate's deposition and mineralization, indicating osteogenic differentiation.
- Surface Roughness Characterization:** Apply scanning electron microscopy (SEM) to visualize scaffold surface roughness.
- Western Blotting:**
 - Detect and quantify osteogenic proteins.

References

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